# NaturArv3D Data Pipeline

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# Introduction

- Compression study somewhat on hold
  - Until more details hashed out
  - Format requirements, data size expectations
- Very helpful input from SNM — *many thanks Mark, Christy, Thomas, Joni!*

# Format Support

| Format | ImageJ/Fiji | Napari | VGSTUDIO | Amira/Avizo | 3DSlicer | Dragonfly | Checkpoint | Morphosource | Data Types | Comment |
|---|---|---|---|---|---|---|---|---|---|---|
| BioFormats | Yes[1] | Yes[1] | No | Yes | No | No | No | No | | |
| TIFF | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | u/i[8/16/32], f16/32 | |
| BigTiff | Yes | Yes[1] | No? | Yes | No? | No? | No? | No? | u/i[8/16/32], f16/32 | |
| JPEG-1 | Yes | Yes | Yes | Yes | Yes | Yes | Yes | Yes | u8 (u16 LL) | |
| JPEG-2000 | Yes | ? | Yes | Yes | No? | No? | No? | Yes? | u1-38 | |
| JPEG-XR | No[2] | ? | No? | No[2] | No? | No? | No? | No? | u8/16/32 | |
| JPEG-LS | No? | ? | No? | No? | No? | No? | No? | No? | u8/16/32 | |
| JPEG-XL | No | Yes[1] | No | No | No | No | No | No | u8/16/32 | |
| DICOM | Yes | Yes[1] | Yes | Yes | Yes | Yes | Yes | Yes | u8/16 | Likely no JXL support? |
| OME-XML | Yes[1] | Yes[1] | No | Yes | No | No | No | No | u/i[8/16/32], f16/32, … | |
| OME-TIFF | Yes[1] | Yes[1] | No | Yes | No? | No? | No? | No | u/i[8/16/32], f16/32, … | |
| OME-ZARR | Yes[1] | Yes[1] | No | Yes[3] | No | No | No | No | u/i[8/16/32], f16/32, … | |
| HDF5 | Yes | Yes[1] | No[4] | Yes[1] | No | Yes? | No | No | u/i[8/16/32], f16/32, … | |
| PNG | Yes | Yes | Yes | Yes | Yes | Yes | No | Yes? | u8/16 | |
| PPM | Yes | ? | Yes | Yes | No? | No? | No | No | u8/16 | |
| RAW | Yes | ? | Yes | Yes | Yes | Yes | No | No? | | |
| BMP | Yes | ? | Yes | Yes | Yes | Yes | No | Yes | u8 | |
| Analyze/HDR | Yes? | Yes[1] | No | Yes | Yes | Yes | Yes | No | u8, i16/32, f32/64, … | |
| NifTI | Yes | Yes[1] | No | Yes | Yes | Yes | Yes | No | u/i[8/16/32], f32/64, … | |
| Other notable | | | | | Many | | FITS (.fits), Kitware VTK (.vtk) | | | |
| Source | | | E-Mail with *Technical Solutions Export* | | 3D Slicer Wiki | Dragonfly Manual | Checkpoint | Duke | | |
| Comment | FOSS (GPLv3) | FOSS (BSD-3), Plugin system | Proprietary | Proprietary | FOSS (BSD), Anything via Python | | Proprietary | Museum data repository | | |

[1] With plugin
[2] Supported through `.czi` format
[3] With plugin by Janelia
[4] Does support HDF4

# Software Support: Summary

- Not widespread support for formats modern formats
  - Wonder how good support would actually be for a 256GB TIFF stack in practice?
- Large datasets have been rare, but is changing
- DICOM format very promising
  - Supports JPEG-LS, JPEG-2000
  - Only widely supported format with lossy compression
  - Even supports JPEG-XL as of 2025
    - Likely implementors do not yet comply
    - Can beat implementors over the head if they state DICOM support
  - Have been very impressed with JPEG-XL
    - Mayo: 50-60% size reductions over existing lossy compressions with no visual difference[1]
- Opinion: ~~Insane~~ Silly to reduce bit-depth to 8-bit but not use (lossy) compression

[1] https://www.dicomstandard.org/news-dir/current/docs/sups/sup232-slides.pdf

# Example: Czeibart, UHR Skulls

Kalman Czeibart, 430 skull CT scans (Christy)

- Source: https://nc.elte.hu/s/doZnwPjgeBaZ7as
- NifTI, gzip, `int16`, $\approx 500 \times 500 \times 1000$
- Sample `001_A_AIRT`: 142MB compressed, 234M uncompressed
  - $\mathrm{CR} = 1.65$, $\mathrm{SS} = 39\%$
- 430 samples = 47.9GB compressed
- Wide software support

**If we have similar size, no need to bother with lossy or non-standard compression!**

# Examples: Beamlines

- EMBL/DESY P14 beamline: TIFF/OME-Zarr/WEBKNOSSOS, see here
- MAX IV: HDF5
- Most beamlines store data uncompressed/losslessy compressed
  - Those that have switched to Zarr likely use Blosc (lossless)

**Even beamlines with huge data storage requirements have not yet adopted lossy compression widely.**

**But note**: they have switched to chunked storage formats, likely out of necessity!

# MorphoSource

- Samples in their CT collections seem to be mostly in 50MB – 2GB range
- Supports formats: TIFF, JPEG-1, BMP, DICOM[1]
  - Not even NifTI support
- Institutions with deposits over 1TB expected to pay
  - Fee is 3.47 USD/GB + 30 USD flat fee[2]
- **Question**: *Is this the intended/expected/desired storage solution for NaturArv3D?*

[1] https://duke.atlassian.net/wiki/spaces/MD/pages/35422892/Critical+Formatting+Details
[2] https://duke.atlassian.net/wiki/spaces/MD/pages/274268722/Data+Deposit+Costs

# Size Considerations

- Detectors:
  - Large samples: Spectrumlogic, $5606 \times 4800$ px, pixel size $50\,\mu\text{m}$
  - Smaller samples: Varex, $3072 \times 3072$ px, pixel size $139\,\mu\text{m}$
- *Optimistic* canonical volume: $3072^3$ vx $= 28.9$ Gvx (assuming 16-bit)
- Assume 50% (?) crop as an average: $14.4$ Gvx
  - 16-bit: $28.8\,\text{GB}$ uncompressed

Cloud storage cost: Backblaze B2 Object Storage[1]:

- Includes 300% monthly egress (data can be downloaded 3 times per month for before additional fees)
- Price is for 5 years of redundant, 'active archive' storage
- $83$ USD / TB / year $\Rightarrow$ $0.415$ USD / GB $(2.65$ DKK$)$
  - Significantly cheaper than MorphoSource, though this is only for 5 years

[1] https://www.backblaze.com/cloud-storage/pricing, accessed 2026-08-26

# Size Consideration Table

| Data / Scheme | *Czeibart* 500×500×1000 0% crop | | 16-bit | $2^3$ × *Czeibart* 1000×1000×2000 0% crop | | 16-bit | ***Small Detector*** 3072×3072×3072 50% crop | | 16-bit | *Large Detector* 5606×5606×4800 50% crop | | 16-bit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Uncompressed | 500 MB | 250 GB | 13.0 TB | 4.00 GB | 2.00 TB | 104 TB | 29.0 GB | 14.5 TB | 756 TB | 151 GB | 75.4 TB | 3.94 PB |
| CR = 1 | 11.1 kr | 5.54 kkr | 289 kkr | 88.7 kr | 44.3 kkr | 2.31 Mkr | 643 kr | 321 kkr | 16.8 Mkr | 3.34 kkr | 1.67 Mkr | 87.3 Mkr |
| SS = 0.00% | 1.33 kr | 663 kr | 34.6 kkr | 10.6 kr | 5.30 kkr | 277 kkr | 76.9 kr | 38.4 kkr | 2.01 Mkr | 400 kr | 200 kkr | 10.4 Mkr |
| **Lossless** | 294 MB | 147 GB | 7.67 TB | 2.35 GB | 1.18 TB | 61.4 TB | 17.1 GB | 8.53 TB | 445 TB | 88.7 GB | 44.4 TB | 2.32 PB |
| CR = 1.7 | 6.52 kr | 3.26 kkr | 170 kkr | 52.2 kr | 26.1 kkr | 1.36 Mkr | 378 kr | 189 kkr | 9.87 Mkr | 1.97 kkr | 984 kkr | 51.3 Mkr |
| SS = 41.17% | 780 mkr | 390 kr | 20.3 kkr | 6.24 kr | 3.12 kkr | 163 kkr | 45.2 kr | 22.6 kkr | 1.18 Mkr | 235 kr | 118 kkr | 6.14 Mkr |
| Lossy | 20.0 MB | 10.0 GB | 522 GB | 160 MB | 80.0 GB | 4.17 TB | 1.16 GB | 580 GB | 30.3 TB | 6.03 GB | 3.02 TB | 157 TB |
| CR = 25 | 443 mkr | 222 kr | 11.6 kkr | 3.55 kr | 1.77 kkr | 92.6 kkr | 25.7 kr | 12.9 kkr | 671 kkr | 134 kr | 66.9 kkr | 3.49 Mkr |
| SS = 96.00% | 53.0 mkr | 26.5 kr | 1.38 kkr | 424 mkr | 212 kr | 11.1 kkr | 3.08 kr | 1.54 kkr | 80.2 kkr | 16.0 kr | 8.00 kkr | 417 kkr |
| **Lossy** | 10.0 MB | 5.00 GB | 261 GB | 80.0 MB | 40.0 GB | 2.09 TB | 580 MB | 290 GB | 15.1 TB | 3.02 GB | 1.51 TB | 78.7 TB |
| CR = 50 | 222 mkr | 111 kr | 5.78 kkr | 1.77 kr | 887 kr | 46.3 kkr | 12.9 kr | 6.43 kkr | 335 kkr | 66.9 kr | 33.4 kkr | 1.75 Mkr |
| SS = 98.00% | 26.5 mkr | 13.3 kr | 692 kr | 212 mkr | 106 kr | 5.53 kkr | 1.54 kr | 769 kr | 40.1 kkr | 8.00 kr | 4.00 kkr | 209 kkr |
| Lossy | 5.00 MB | 2.50 GB | 130 GB | 40.0 MB | 20.0 GB | 1.04 TB | 290 MB | 145 GB | 7.56 TB | 1.51 GB | 754 GB | 39.4 TB |
| CR = 100 | 111 mkr | 55.4 kr | 2.89 kkr | 887 mkr | 443 kr | 23.1 kkr | 6.43 kr | 3.21 kkr | 168 kkr | 33.4 kr | 16.7 kkr | 873 kkr |
| SS = 99.00% | 13.3 mkr | 6.63 kr | 346 kr | 106 mkr | 53.0 kr | 2.77 kkr | 769 mkr | 384 kr | 20.1 kkr | 4.00 kr | 2.00 kkr | 104 kkr |

- Cell rows: data size, storage cost (MorphSource), storage cost (Backblaze B2)
- Cell columns: 1 recon., 1 week of recons. (shaded, 500), 1 year of recons. (26089)
- Units have been SI-ified: mkr = milli-kroner, Mkr = mega-kroner/million, etc.

# Potential Storage Solutions

- Lowest Common Denominator
  - Appears to be uncompressed TIFF/DICOM/JPEG-1/PNG
  - Note: Likely that many of them don’t support JPEG-XL in DICOM yet?
- Zarr + Objects (+ Local Conversion)
  - Build or find small tool that can convert Zarr from object storage to TIFF locally
- A Reasonably Common Denominator
  - HDF5 (like MAX IV and others, likely still spotty codec support)
- Zarr/OME-Zarr/TIFF/… + REST API
- …?

# The Case for Zarr

- Zarr has seen rapid adoption
  - NASA is commited — VEDA/ODD team
  - OME-Zarr is leading NGFF (Next Generation File Format) selected by OME
  - EMBL is adopter (DESY beamlines, Image Data Resource (IDR))
  - Janelia is adopter (OpenOrganelle, other projects)
  - + Many other prestigious and important institutions
- Zarr is vastly technologically superior to legacy formats
  - Access: Chunking, parallel access, multi-dimensional, cloud friendly (object storage)
  - Efficiency: **Codecs**, streaming/async, performant
  - Open: **Open standard, open governance**, highly interoperable
- Opinion: **Zarr/OME-Zarr is the correct long-term solution**
  - *May not be acceptable here due to legacy software support and data repository requirements*

# Extra: JPEG-XL Lossy from Last Meeting

- 3D reconstructions take up bulk of storage
- Can be remade from archived projections
- This enables lossy compression
- Lossy JPEG-XL:
  - Distance 1 (middle row) gives 98.9% space saving
    - CR=90.5, SSIM=0.9992
  - Distances >1 produce noticeable artefacts and smoothening
  - Distance 0.1 gives 92.1% space saving
    - CR=12.7, SSIM=0.9995
- Demonstrates how good lossy compression is
  - mostly for SNM's benefit

